

Grouping the Harvard EEG reports with doc2vec

We wanted to see whether the MGH EEG reports fall into natural groups, and in particular whether reports written by the same neurologist end up near each other. This is the doc2vec idea from Alexander's Fraser Health code, run on the Harvard reports.

Each of the 55,557 MGH reports was turned into a 100-dimensional vector with doc2vec (the same settings as Alexander's — window 5, 100 epochs, min_count 3, PV-DM) and then projected down to two dimensions with UMAP. Where the de-identification happened to leave a reader's initials in the signature line, we tagged the report with that neurologist; that gave us eight neurologists with at least forty reports each. We tried it both ways: with the signature cut out, so only the writing itself counts, and with it left in.

Plotted plain, the reports look like this:

MGH EEG reports (n=55,557) — Doc2Vec → UMAP

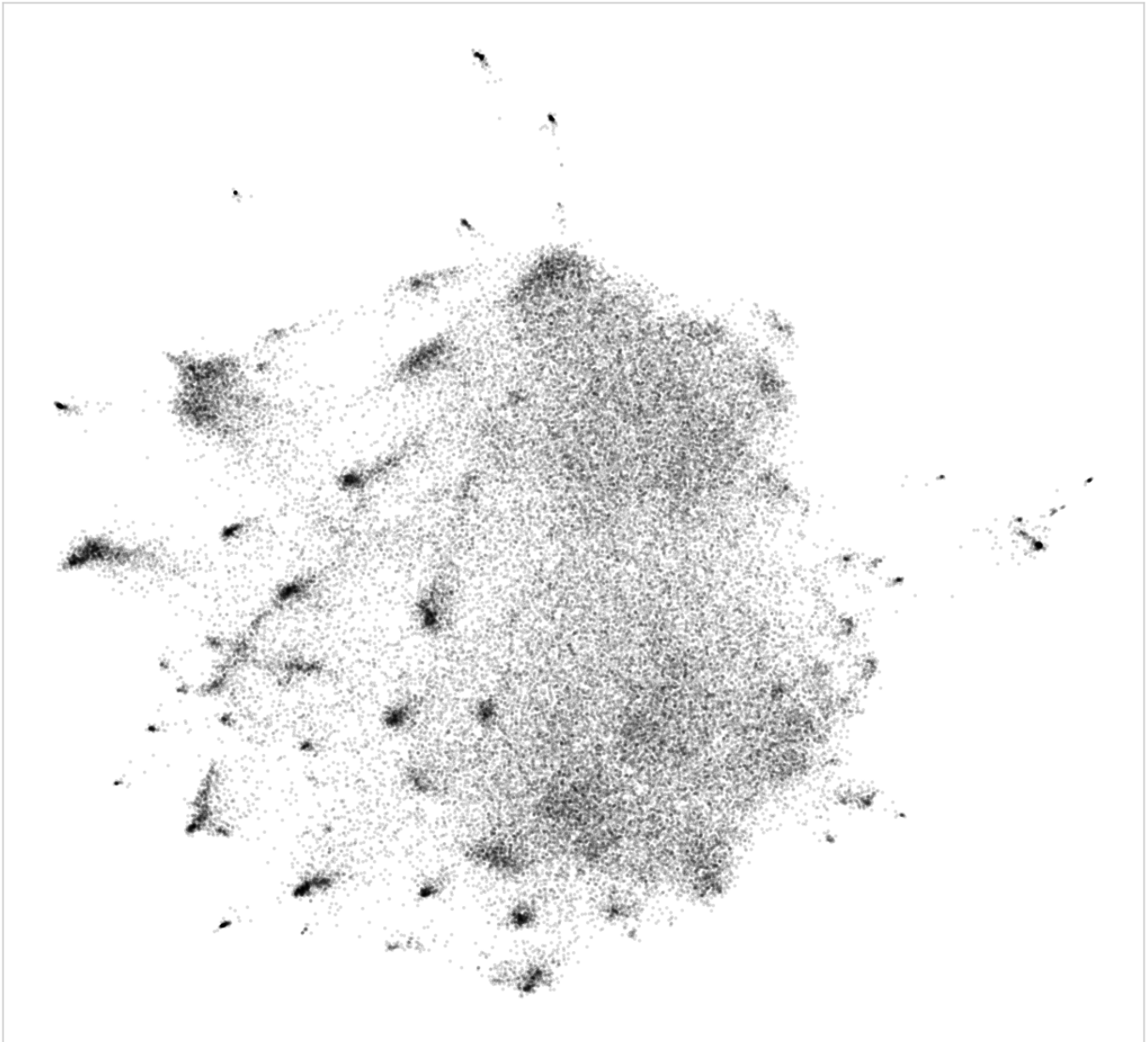


Figure 1: All MGH reports

One big loose cloud with a scatter of small, tight knots around it. Those knots turn out to be near-identical templated reports — the standard “Normal EEG” text and the like — not anything to do with who wrote them.

Coloring the eight neurologists on the same map:

Do the reports cluster? all MGH reports (55,557)

Doc2Vec (100-d, signature kept) → UMAP · silhouette -0.01 · 5-NN author accuracy 53% vs 44% baseline

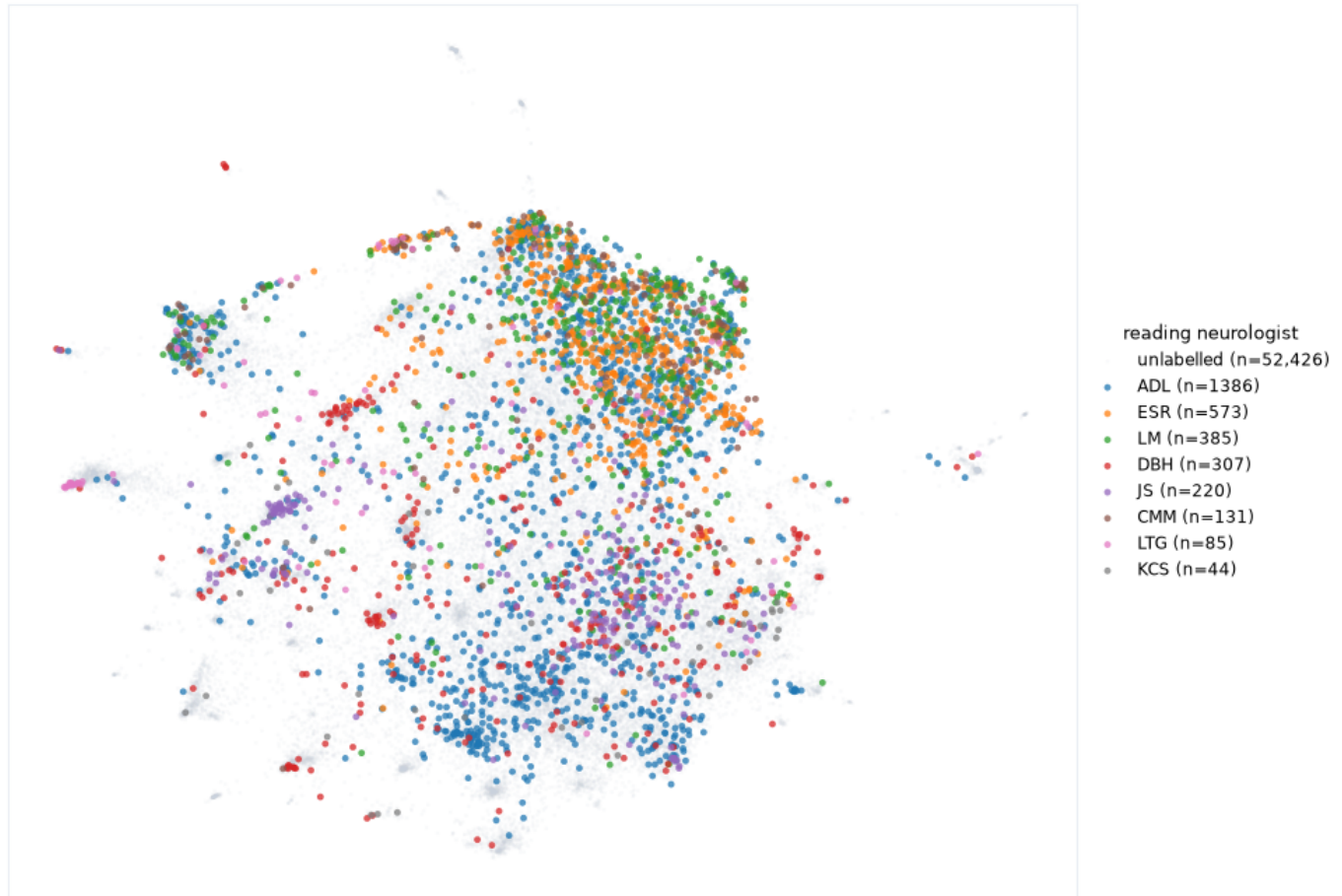


Figure 2: Colored by neurologist

The colors are spread right through the cloud. A couple of readers lean to one side (ESR a bit higher, ADL a bit lower), but nobody forms their own group. The numbers agree: however we measure it, telling the neurologist from the report is barely better than guessing the most common one — about 47–53% against a 44% baseline, and a silhouette score near zero. Clustering the whole map lumps 86% of the reports into a single group.

So with doc2vec the reports don’t sort by neurologist, with or without the signature in the text. The grouping that is there is about the report template, not the person — most likely because these reports are so templated that the shared boilerplate, which doc2vec averages over the whole document, drowns out any individual style. If we want to keep after the “who wrote it” question, the next thing worth trying is character n-grams after stripping out the boilerplate.

To reproduce:

```
python -m analysis.harvard_doc2vec_authors # signature removed
STRIP_SIG=0 python -m analysis.harvard_doc2vec_authors # signature kept
```